Question 1:

With the help of truth table, verify whether the following statement (S) is a tautology or no.

a. ***[3 Points]*** S1: $[(P \vee Q) \wedge (\neg P \vee R)] \rightarrow (Q \vee R)$

b. **[2 Points]** S2: $(P \wedge (P \rightarrow Q)) \rightarrow Q$ **What is this also called?**

c. ***[4 Points]*** Convert the following statement to CNF and DNF both: $(P \leftrightarrow (Q \vee R)) \wedge (\neg S \rightarrow T)$

$Q \vee R$

Question 2:

a. [2.5 Points] I am storing a bit on my hard drive. I have changed my OS to return it in different way if a guest user tries to access it. This program simulates a fair coin toss and for Head, returns the original bit. If Tail, it again simulates a fair coin toss and if Head this time, returns 1 else 0. If probability of my original bit, say x, is one is p. That is, $Pr(x=1)=p$. If you login to my computer using guest account and aware of the above program. If you read the bit to be 1, what is probability that the original bit is 1?

b. [3 Points] Let the events be as
   i. S: Proposition that patient has stiff neck
   ii. H: Proposition that patient has severe headache
   iii. M: Proposition that patient has meningitis

Assume S and H are independent given the patient has M.

Prove the following statement:

$$P(M \mid S \wedge H) = P(S \mid M) * P(H \mid M) * P(M) / P(S \wedge H)$$

c. A certain disease affects 1% of the population. A test for the disease is 99% accurate, meaning it correctly identifies 99% of those with the disease (true positive rate) and correctly identifies 99% of those without the disease (true negative rate).
   i. [2 Points] If a person tests positive, what is the probability that they actually have the disease?
   ii. [2.5 Points] What is Accuracy, Precision and the Recall of the above test?

d. [3 Points] Suppose you have 6 data points in 2d with lables +1 or -1 as: (1,1), (2,2) and (2,0) belonging to +1 and (5,5),(5,6) and (6,5) belonging to the class -1. You received a new point (3,3) and you want to determine its label using kNN with k=1 and k=3. What would be the labels respectively? Draw 2-dimensionaldigram and mark the above points. Use Euclidian distances.

Question 3:

[12 Points] Consider the following data (as seen in the class)

5 of 5 columns

Detail Compact Column

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny 5 36% 3,2	Mild 6 ~~4~~ 43% 4,2	7 High 4,3	Weak 8 57% 6,2	true 9 64%
Rain 5 36% 3,2	Hot 4 29% 2,2	7 2 unique values 6,1	Strong 6 43% 3,3	false 5 36%
Other (4) 4 29% 4,0	Other (4) 4 29% 3,1			

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

We need to make a decision tree to decide whether to play tennis or no. We would use CART algorithm step to split the tree. Here, splitting happens on the attribute which has least Ginni impurity rather than maximum information gain.

$GI(S,A) = \sum_t \frac{S_t}{S} GI(S_t)$, where,

i. $t \in T$ is the different values present in S for A,

ii. $GI(S_t) = 1 - \sum_{i \in K} p_i^2$, K is the number of classes and p_i is probability of a random sample in class i. (we estimate it from the data, as we did it the class)

Derive Decision tree for the above data.

Question 4:

3.96

a. **[2 Points]** Consider the following transaction table.
From this, Mr X has made the following Association Rule (AR)
AR: : Tomato, Potato → Onion
What is support, confidence and lift of the above rule?

Transaction id	Items
1	Tomato, Potato, Onion
2	Tomato, Potato, Brinjal, Pumpkin
3	Tomato, Potato, Onion, Chilly,
4	Tamarind, Lemmon
5	Tomato, Potato, Tamarind, Chilly, Onion
6	Tamarind, Chilly, Onion

0.66

b. **[1 Points]** In an ϵ greedy strategy for a multi-armed bandit, what is the main purpose of choosing a random arm with probability ϵ?

c. **[1 Points]** In Multi-Armed Bandit settings, what is the probability that $\epsilon-$greedy algorithm incurs $O(T)$ regret if there are two arms having rewards drawn from bernoulli trials with mean 1 and 0. Provide small justification.

d. **[1 Points]** Consider a 3-armed bandit problem with unknown reward means. Which approach best describes how Upper Confidence Bound (UCB1) selects an arm at each time step?

Circle the correct answer. (1M) (wrong answer incurs penalty of -1M)

a. It selects the arm with the lowest number of pulls to maximize exploration

b. It selects the arm that maximizes a sum of its empirical mean reward and an exploration bonus that shrinks as the arm is pulled more

c. It selects the arm with the highest current empirical mean reward

d. It selects arms uniformly at random until each arm has been pulled the same number of times, then exploits the best arm forever

1

e. **[6 Points]** Consider single pull multi-armed bandit setting with 3 arms and 10 rounds. Suppose the following table, the row t indicates the reward if you had pulled the corresponding arm in round t. Can you explain how UCB1 would have pulled the arms from t=1to 10. (Assume that UCB1 does only once round robin in the beginning and then follows UCB)

Round \ Arm	1	2	3
1	1	0	1
2	0	1	0
3	1	1	0
4	0	0	1
5	1	1	1

6	0	1	0
7	1	0	1
8	0	1	1
9	1	0	0
10	1	1	1

Question 5:

[6 points] A six-sided fair dice is rolled 10 times and then 20 times. The outcomes are: [5,3,6,4,5,1,1,4,2,3] and [1,4,3,2,2,3,6,5,3,2,4,1,3,6,2,5,3,4,2,6]. A machine learning model estimates the probability of each face as the count of each face divided by the total number of rolls. Please compute the bias, variance and MSE of the ML model for each of the (two) set of rolls. Please present all the steps involved.

Question 6:

Consider a Markov Decision Process (MDP) with 4 states {S1, S2, S3, S4} arranged in a row from Left to Right with a start state of S1 and terminal state S4 with +~~100~~ 70 reward. There are 4 actions for the agent {Up, Down, Left, Right}. When an agent takes action, it goes one step in the corresponding direction with probability p and perpendicular directions with (1-p)/2. When the robot hits a wall, it gets back to the same state. Step cost is SC and discount factor is DF. Please assume p is 0.8 unless stated otherwise. Let the optimal policy for the Robot be as follows:

S1	S2	S3	S4
→ (Right)	→ (Right)	→ (Right)	+70

Instead of computing the policy, an ML algorithm was used to train the policy and it learnt a policy as follows:

S1	S2	S3	S4
→ (Right)	↑(Up)	→ (Right)	+70

Please compute the following:

a. **[4 points] Case 1:** What is the Expected Utility for each of the 4 states for this Robot using Optimal policy and what is the Expected Utility using the ML policy ? Please use SC = 0 and DF = 1.

b. **[4 points] Case 2:** What is the Expected Utility for each of the 4 states for this Robot using Optimal policy and what is the Expected Utility using the ML policy ? Please use SC = -2 and DF = 0.9.

c. **[3 points]** For each of the Cases 1 and 2, can you compute MSE for the policy identified by the ML model ? Please provide steps of the computation if yes or the detailed reasoning for why not able to ?

d. **[4 points]** Please compute the A matrix for the above MDP (that would be needed as part of using a Linear Programming approach).

Question 7:

The popular cricketing site Cricinfo provides the winning chance of the teams (T1, T2) at any point of the game as (z%, 100-z%). Please answer the following based on this:

a) **[3 points]** You took a bet that T1 would win. Both winning and losing the bet would be same value of 300 rupees (i.e., +300 or -300). Please mention whether the following statement is True or False and present the calculations to support: You would expect to receive 180 rupees if z is 60%.

2 60

b) **[3 points]** If you are a risk neutral agent, would you play the bet in (a)? Please mention True or False and present a suitable utility function to make the case for your answer along with expected utility calculations involved.

c) **[3 points]** If you are a risk averse agent, you would definitely not play the bet in (a). Please mention True or False and present a suitable utility function to make the case for your answer along with expected utility calculations involved.

d) **[3 points]** You are a risk neutral agent. While the current status from Cricinfo is (60%, 40%), another Cricketing Site CS which you looked through predicts the game as (40%, 60%) at the same time. You trust Cricinfo by 60% and CS by 40%. Given this what would your bet be if (winning, losing) would result in (+300, -400) rupees. Please present a suitable utility function to make the case for your answer along with the expected utility calculations involved.

$f(x) = x$

expected winning

Question 8: Please answer the following questions:

a) **[4 points]** Two friends are planning to go for a movie. Friend F1 prefers movie M1 (i.e., obtains utility of 1) while friend F2 prefers movie M2 (i.e., utility of 1). F1 and F2 would obtain 0 utility if the choices are swapped. From F1's perspective, F2 is using arguments that move the outcome towards (a target) M2 which can be viewed as a form of Regularization (say R). Please note F2 is a friend whose arguments you cannot ignore, hence need to account in proportion to how persuasive or strong the arguments are (reflection of how we act in real life). How do we model the ML objective for F1 without R and what would be the suitable R to use ? Please then present what the final objective function would be and what it would do without and with R so the difference is clear (and gradual to capture the strength of argument).

b) **[3 points]** What is feature engineering? Please explain clearly how the two techniques namely Scaling and Binning can be used to perform feature engineering?

Question 9:

[6 points] DFS is considered to be memory efficient compared to BFS. Please consider a binary tree which has all nodes to depth 4 where root node is at Depth 0 (D0) i.e., tree has 1, 2, 4, 8, 16 nodes (at D0, D1, D2, D3 and D4). Please consider tree search version for both and when needing to make a choice at a node, left node is preferred over right node for expansion. You are given choice to pick one goal node which is same for both algorithms. Please pick a goal node such that the step needing the maximum memory in BFS (smmBFS) minus the step needing maximum memory in DFS (smmDFS) [i.e., smmBFS - smmDFS] is highest. Please mention the goal node picked, illustrate the nodes in memory for smmBFS and smmDFS and present the maximum difference involved. Please provide clear reasoning for why you believe this would be the maximum value.

Question 10:

A* graph search is optimally efficient i.e., finds optimal solution by expanding fewest number of nodes. A robot is navigating a 4*4 grid world where {rows, columns} are numbered 1 to 4 from {top to bottom, left to right} resulting in 16 cells. The bottom ~~right~~ left point is at origin (0,0) (i.e., entire grid lies in the first quadrant). Each cell is of dimension 2*2 (hence cells go from 0 to 8 on both x and y axes). Robot is in cell (1,3) and goal state is cell (4,1). At each time step, Robot is allowed to take one step in the direction of {Up, Down, Left, Right} (only if a valid cell is present) whose step cost would be {0,3,2,1} units. Robot is assumed to start at center point of a cell and action takes it to the center point of a new cell, hence covers physical distance of 2 units per step (no other position within a cell is allowed).

- **[3 points]** Please define a heuristic function and provide clear reasoning that it satisfies Admissibility and Consistency properties. Please note that the heuristic will need to satisfy geometric properties of the grid which lies on a plane surface.

- **[3 points]** Assuming such a heuristic is designed, please present all steps of working of the A* graph search algorithm. Please mention how many node expansions were needed to find the solution.

- **[2 points]** Using the heuristic designed, please present all steps of working of Best first graph search algorithm for the information presented. Please mention how many node expansions were needed to find the solution.

- **[1 points]** Based on the above computations, please argue whether or not A* graph search is optimally efficient.